

Project Title	Project Title: U.S. Airline Performance & Delay Analysis 
Skills take away From This Project	Welcome to your capstone project! This project challenges you to analyze a large dataset of U.S. domestic flights to uncover patterns in delays and cancellations, evaluate airline and airport performance, and provide data-driven insights. • Advanced SQL Querying (joins, aggregations, filtering complex data) • Business Intelligence with Power BI/Tableau (dashboard creation, data storytelling) • Key Performance Indicator (KPI) Development & Analysis • Data Cleaning, Transformation, and Integration • Exploratory Data Analysis (EDA) • Performance Benchmarking • Analytical Report Writing & Presentation
Domain	Aviation Analytics / Transportation Management

Problem Statement:

Flight delays and cancellations are significant issues in the U.S. aviation industry, impacting passengers, airlines, and the economy. Your task is to perform an in-depth analysis of historical flight data to identify key drivers of these disruptions, assess performance, and propose actionable insights.

Business Use Cases:

The project's solution can be implemented in various scenarios, including:

- Analyzing the primary causes and patterns of flight delays and cancellations for strategic operational adjustments.
- Benchmarking the on-time performance, delay severity, and cancellation rates of different airlines for competitive analysis and improvement initiatives.
- Evaluating the operational performance of various U.S. airports to identify bottlenecks and areas for infrastructure or process improvement.

- Investigating how factors like time of day, day of week, month, and specific routes affect flight operations to optimize scheduling and resource allocation.
- Providing data-driven recommendations to stakeholders (airlines, airports, regulatory bodies) to enhance passenger experience and operational efficiency.

Approach:

Phase 1: Project Setup & Data Ingestion

- Set up your SQL database environment.
- Create appropriate table structures for flights, airlines, and airports data. Consider data types carefully.
- Import the data from the CSV files into your SQL database.
- Perform initial verification (e.g., row counts, sample data review) to ensure data is loaded correctly.

Phase 2: Data Cleaning, Preparation & Integration (SQL)

- **Time/Date Handling:** Convert text-based time and date fields (e.g., SCHEDULED_DEPARTURE) into proper datetime formats. This may involve creating new columns. Think about how to combine year, month, day, and HHMM time strings.
- **Missing Values:** Investigate and handle missing values (NULLs) in key columns (e.g., delay columns, cancellation reason). Decide on an appropriate strategy (e.g., replace with 0 where logical, leave as NULL, or flag).
- **Data Enrichment:**
 - Create a more descriptive CANCELLATION_REASON_DESC column based on the CANCELLATION_REASON code.
 - Consider creating a FLIGHT_DATE column for easier time-based analysis.
- **Integration:** Create a unified analytical dataset (e.g., a comprehensive SQL View or a new table) by joining flights data with airlines and airports (for both origin and destination) to include descriptive names and locations.

Phase 3: Exploratory Data Analysis (EDA) & KPI Definition (SQL)

- Using SQL queries on your integrated dataset, explore:
 - Overall flight volumes, cancellations (total, by reason), and diversions.
 - Basic statistics for departure and arrival delays (average, median, min, max).
 - The distribution of different types of delays (airline, weather, NAS, etc.).
- **Define Key Performance Indicators (KPIs)** relevant to the project objectives. Examples include:
 - On-Time Performance (OTP) Rate (e.g., flights arriving within 15 minutes of schedule).

- Average Arrival/Departure Delay (in minutes).
- Cancellation Rate (%).
- Percentage contribution of each delay type.
- Perform initial aggregations to understand how these KPIs vary by:
 - Airline
 - Origin/Destination Airport
 - Month, Day of Week, Time of Day (requires robust time parsing)

Phase 4: Dashboard Development (Power BI / Tableau)

- Connect your BI tool to your SQL database (or the integrated view/table).
- **Data Modeling in BI Tool:**
 - Define necessary relationships if you load multiple tables.
 - Create calculated measures/fields required for your KPIs (e.g., Total Flights, OTP Rate, Avg Delay, Cancellation Rate). *You will need to translate your KPI definitions into the BI tool's formula language (DAX for Power BI, or Tableau's functions).*
- **Design and build your interactive dashboard. Consider including pages/views for:**
 - **Overview:** Key KPIs, overall delay/cancellation reasons.
 - **Airline Performance:** Comparative analysis of airlines against KPIs.
 - **Airport Performance:** Comparative analysis of airports; consider a map visualization.
 - **Temporal Trends:** How KPIs change by month, day of week, or hour.
- Ensure your dashboard includes filters/slicers for interactive exploration (e.g., by airline, airport, date).

Phase 5: Insight Generation & Recommendation Formulation

- Analyze your dashboards and SQL query results deeply. Ask "why?" and "so what?".
- Identify significant trends, patterns, anomalies, and correlations.
- Based on your findings, formulate 2-3 actionable and data-driven recommendations for potential stakeholders (e.g., airlines, airport authorities). Consider what changes could realistically lead to improvements.

Phase 6: Reporting & Presentation

- **Final Report:** Structure your report logically (Introduction, Methodology, Key Findings & Analysis, Dashboard Overview, Recommendations, Conclusion). Use visuals from your dashboard to support your findings.

- **Presentation:** Prepare a concise and engaging presentation summarizing your project, highlighting key insights, and showcasing your dashboard.

Results:

- Analyze the primary causes and patterns of flight delays and cancellations.
- Benchmark the on-time performance, delay severity, and cancellation rates of different airlines.
- Evaluate the operational performance of various U.S. airports.
- Investigate how factors like time of day, day of week, month, and route affect flight operations.
- Translate your findings into meaningful recommendations for stakeholders.

Project Evaluation metrics:

- **SQL Proficiency:** Correctness, efficiency, and clarity of your SQL scripts.
- **Data Handling:** Accuracy and completeness of your data cleaning and preparation.
- **Dashboard Effectiveness:** Clarity, usability, insightfulness, and interactivity of your Power BI/Tableau dashboards.
- **Analytical Depth:** Actionability and depth of the insights derived from your analysis.
- **Report & Presentation Quality:** Professionalism, clarity, and comprehensiveness of your final deliverables.

Technical Tags:

- **Primary:** SQL, Power BI (or Tableau), Microsoft Excel (for supplementary tasks).
- **Minimal/Optional:** Python (e.g., for initial bulk data handling if preferred, but not a core requirement for analysis).

Data Set:

- **Source:** Google Drive: [2015 Flight Delays and Cancellations](#)
- **Format:** CSV (Comma Separated Values)

Data Set Explanation:

- **Content:** This dataset contains detailed records of domestic flights in the USA for 2015, including scheduled/actual times, delays, cancellations, airlines, and airports.
- **Key Files:**

- flights.csv: Main flight transaction data.
- airlines.csv: Lookup table for airline names.
- airports.csv: Lookup table for airport details (including location).
- **Important Columns to Explore (in flights.csv):** YEAR, MONTH, DAY, DAY_OF_WEEK, AIRLINE, FLIGHT_NUMBER, ORIGIN_AIRPORT, DESTINATION_AIRPORT, SCHEDULED_DEPARTURE, DEPARTURE_TIME, DEPARTURE_DELAY, SCHEDULED_ARRIVAL, ARRIVAL_TIME, ARRIVAL_DELAY, CANCELLED, CANCELLATION_REASON, AIR_SYSTEM_DELAY, SECURITY_DELAY, AIRLINE_DELAY, LATE_AIRCRAFT_DELAY, WEATHER_DELAY, DISTANCE.

Key preprocessing steps required include:

- **Handling Time and Date:** Converting string representations of dates and times (e.g., "0855" for HHMM) into proper datetime formats to enable temporal analysis. This often involves combining year, month, day, and time components.
- **Managing Missing Values:** Addressing NULLs in critical columns such as delay durations (which might be NULL if not delayed or if cancelled) and cancellation reasons.
- **Data Type Conversion:** Ensuring all columns are of appropriate data types for analysis.
- **Data Enrichment:** Creating new descriptive columns, for example, mapping CANCELLATION_REASON codes (A, B, C, D) to their textual descriptions (Airline/Carrier, Weather, etc.).
- **Data Integration:** Joining the flights.csv data with airlines.csv and airports.csv to incorporate descriptive names for airlines and airports (origin and destination), as well as airport geographical information (city, state, latitude, longitude) into a unified analytical dataset or view.

Project Deliverables:

1. **SQL Scripts Package:** Well-commented SQL scripts used for data cleaning, transformation, integration, and all analytical queries performed to derive insights.
2. **Interactive Dashboard (Power BI/Tableau):** The BI tool file (e.g., .pbix for Power BI, .twbx for Tableau) containing a multi-page dashboard that visually presents the analysis, key findings, and allows for interactive exploration.
3. **Final Project Report (Word/PDF Document):** A comprehensive document detailing the project's introduction (problem statement, objectives), methodology (data sources, tools, analytical approach), key findings and analysis (supported by charts/tables), a walkthrough of the dashboard, actionable recommendations, and a conclusion with any limitations.
4. **Presentation (PowerPoint/Google Slides):** A concise slide deck summarizing the project's objectives, methodology, most impactful findings, dashboard highlights, and key recommendations, suitable for presenting to stakeholders.

5. Recorded Video

Project Guidelines:

- Write clean, well-commented, and reasonably efficient SQL code.
- Clearly document all data cleaning and transformation decisions and steps.
- Design dashboards that are intuitive, informative, visually appealing, and effectively tell a story with the data.
- Focus on deriving actionable recommendations that are clearly supported by your data analysis.
- Ensure data consistency and validation at various stages of processing and analysis.
- Use version control (e.g., Git) for managing SQL scripts and documentation if you are comfortable with it, though not mandatory.
- Manage your time effectively across the different project phases to meet deadlines.

Timeline:

The project is expected to be completed following a phased approach.

- **Phase 1: Project Setup & Data Ingestion** (e.g., 1-2 Days)
- **Phase 2: Data Cleaning, Preparation & Integration** (e.g., 1-2 Days)
- **Phase 3: Exploratory Data Analysis (EDA) & KPI Definition** (e.g., 2 Days)
- **Phase 4: Dashboard Development & Refinement** (e.g., 2 Days)
- **Phase 5: Insight Generation & Recommendation Formulation** (e.g., 1 Day)
- **Phase 6: Final Report Writing & Presentation Preparation** (e.g., 1 Day)

Deadline → 7 days